K832 — EOD SIGN-OFF, 2026-07-22 (Keeper). The weak-chirality + charge arc: CHARGE SECTOR DERIVED (bankable, close it); PARITY reduced to ONE well-posed computation (the area is one step from fully closed). Ledger, verdict PASS, tomorrow's plan.

Keeper | 2026-07-22 EOD | Formal audit close of a long, disciplined day (K821–K832; toys 4781–4791; ~24 team notes). Verdict below. No false thing banked across ~11 refuted closures.

THE DAY'S LEDGER — what banks, what's located

? DERIVED / banked (survived the arithmetic)

- CHARGE SECTOR — DERIVED (given the SM reps), non-circular. Anomaly-freedom + the Z_6 center correlation 6Y≡4t+3d (mod 6) force the SM hypercharges + exact quark/lepton charges {+2/3,−1/3,−1,0}. The correlation SPLITS into color (4t = T2521's 1/N_c, banked target-innocent) + isospin (3d = U(2) Kähler holonomy, K813) → non-circular on the banked T2521. Grace's N_c-neutrality (T2521) FIRMED from imposed → DERIVED via anomaly-inflow. (K828/K829.)
- DIRAC (K822) — signature-forced, over-determined. Bulk fermion is 4D vector-like.
- THE SQUEEZE (K825) — the bulk single spinor is structurally vector-like (Witten no-go in one-manifold Clifford algebra). Numerically verified.
- Non-orientable Shilov boundary (K826) — $(S^4{\times}S^1)/Z_2$ non-orientable → Pin → permits chirality. Verified ×3.
- Which SU(2) = internal isospin (K824); gravi-weak literal form ruled out.
- The grading MECHANISM realized (K830): internal SU(2)_L instanton bundle over S^4=SO(5)/SO(4), c_2 = k = ±1 (computed). The chiral grading exists.
- Custodial/ρ≈1/no-W_R (T2520); CP-free; chirality mechanism theorem (K819) — carried forward.

? LOCATED FRONTIER — parity, reduced to ONE computation

Parity is one well-posed computation from DERIVED. Every upstream link is a checked fact: bulk vector-like (squeeze) → chirality must be a boundary phenomenon → boundary non-orientable (permits) → internal instanton k=±1 (grading exists) → **the chirality is carried by U(1)_Y** (the SU(2) doublet is pseudoreal → vector-like alone; hypercharge Y≠0 makes (2)_Y complex → chiral). **THE SOLE OPEN DECIDER (pure D_IV^5 linear algebra): do the Z_2-projected, instanton-twisted holomorphic sections carry the U(1)_Y charge that makes their rep COMPLEX (→ chiral → parity derived) or a REAL/pseudoreal rep (→ vector-like → derived-conditional)?** Lyra's sharpening: U(1)_Y bundles over S^4 are topologically trivial, so Y is a GLOBAL LABEL on the boundary zero mode — the residual is sharp: what Y (0 or 1/6) does the projection assign the surviving sections? Elie's harness ready. Held at LEAD (11th candidate closure — compute, don't assert).

★ THE UNIFICATION (the day's conceptual headline)

Parity and charge are the SAME U(1)_Y mechanism. The hypercharge that closed the charge leg is exactly what makes the fermions chiral (breaks the SU(2)-doublet pseudoreality). Two legs tracked separately all arc → one object. IF the projection carries Y, parity follows FROM the derived charge sector — a two-way lock.

? REFUTED this arc (~11 closures, each killed by computation — the discipline record)

SO(2)-does-everything · Y=SO(2)=J · Y=broken-SU(2)_R · Lyra F640 Weyl · Elie 4781 Weyl · Grace ω-lock · Route-1 gravi-weak coincidence · naive-orbifold=γ^5 · anomaly-alone→SM (Lyra dropped root) · "k=1→one generation" (Witten/squeeze obstruction) · [held] "Y makes it chiral" (pending the projection computation). Two were mine-adjacent (F277/F279 glueball-scope over-reach; the KK over-import Casey corrected). Nothing false banked.

VERDICT: PASS.

The day's derived results are referee-defensible and honestly tiered. The charge sector is a genuinely closed leg. The parity frontier is precisely located (not a wall, not a hand-wave) — one linear-algebra computation from derived. The consolidation paper (`BST_Weak_And_Charge_Sector_Consolidated_State_2026-07-22.md`) is current and referee-facing.

CLOSING ASSESSMENT (Casey's question: close the area or more?)

- CLOSE the CHARGE sector — DONE, derived, banked.
- PARITY: one computation from closed. Recommend: finish it first thing tomorrow (warm, one step, high value — could derive parity AND lock it to the charge sector). If it lands → the whole EW chirality+charge area closes. If it's genuinely hard → bank the located-frontier (parity mechanism grounded on derived pieces, one link named) and pivot.

TOMORROW

1. PRIMARY (warm, finish the area): the parity last-link — does the Z_2-projection assign the surviving boundary sections nonzero U(1)_Y (→ complex → chiral → parity derived)? Lyra's projection/reduction + Elie's mod-2 index + anomaly-inflow. ONE computation.
2. THEN / ALTERNATIVE fresh non-derived row (Keeper's recommendation): color confinement dynamics — complete the Shilov-vanishing theorem (K744: "confinement DERIVED pending the color-K-type check") to the confinement mechanism. Discrete/structural (BST-strong, the arc's proven lane), adjacent to the charge sector's color center Z_{N_c}, and genuinely un-closed. (Alternative: the neutrino absolute mass scale / Weinberg operator — open, seesaw-discrete.) NOT $\sin^2\theta$_W or quark-mass values — standing honest-negatives (runners/swamp); don't re-open.

— Keeper K832, EOD 2026-07-22. Charge sector DERIVED (given reps, non-circular on T2521); parity reduced to ONE linear-algebra computation (does the Z_2-projection carry U(1)_Y → complex/chiral?); parity+charge = one U(1)_Y mechanism (the unification). ~11 closures refuted, nothing false banked. VERDICT PASS. Tomorrow: finish parity last-link (warm); then color-confinement dynamics (fresh discrete row). See [[Keeper_K831_WITTEN_KK_OBSTRUCTION_coset_fermions_are_vector_like_even_with_nonzero_index_k1_does_NOT_give
07-22]], [[BST_Weak_And_Charge_Sector_Consolidated_State_2026-07-22]], K822/K824/K825/K826/K828/K829/K830.